Outpatient Prototype Scorecard Spreadsheet Review

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On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. Changes shown in square brackets [].

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Some roof u-values for early vintages incorrect (OUTP26-Envelope)
- For 2008-2018 thermal efficiency should be used rather than combustion (OUTP33-HVAC System)
- Ventilation rate incorrect for operating rooms (OUTP41-Ventilation)
- Elevator lighting shutoff timing incorrect (OUTP62-Equipment)
- One weekday office occupancy schedule value is incorrect (OUTP65-Schedule)
- Office equipment schedule for Saturday incorrect (OUTP66-Schedule)

The following are the highest priority comments:

- The prototype is based on one actual surgical center (OUTP13-Prototype)
- Additional air handlers for special spaces are expected (OUTP17-Zones)
- Surgical center may not be most representative of outpatient (OUTP18-Zones)
- Operating room ventilation rate half what is required (OUTP41-Ventilation)
- Ventilation driven by exhaust not computed correctly (OUTP42-Ventilation)
- Floor levels assigned to schedules reversed (OUTP64-Schedule)

The following comments represent other important issues found:

- Furnace efficiency for existing seems too low (OUTP32-HVAC System)
- Operating rooms and imaging usually cooler (OUTP37-Thermostat)
- Additional lighting power is ignored (OUTP48-Interior Lights)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- No justification is provided for the overall floor area (OUTP5-Notes) [moved from other important issues]
- Little justification is provided for major assumptions (OUTP8-Notes) [moved from highest priority]
- The distribution of areas is not justified (OUTP19-Zones) [added to this list]
- Occupant density for healthcare zones not explained (OUTP20-Zones) [moved from highest priority]
- No source for locker room people density (OUTP23-Zones) [added to this list]
- No source for types constructions (OUTP25-Envelope) [added to this list]
- No basis for ratio of single to double pane is provided for early vintages (OUTP28-Envelope) [moved from highest priority]

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Int Lighting
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights
- PV
- EV Charger

The comments on these tabs are labeled using the following format OUTPx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and

<CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the schedules tab.

OUTP1-Notes (n/a)

Rather than repeat it, please see comment PPD1 in my single-family residential prototype review, LHSC1-Notes in the large hotel review, and SHSC1-Notes in the small hotel review.

OUTP2-Notes (n/a)

Rather than repeat it, please see comment PPD2 in my single-family residential prototype review, LHSC2-Notes in the large hotel review, and SHSC2-Notes in the small hotel review.

OUTP3-Notes (n/a)

To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, 32 resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the hospital prototype are the following:

- *ASHRAE Handbook HVAC Applications Chapter 9 "Health Care Facilities"*
  Essential for understanding healthcare HVAC. If only one thing is reviewed, this should be it.
  https://store.accuristech.com/standards/2023-ashrae-handbook-hvac-applications-i-p?product_id=2225673
- *ANSI/ASHRAE/ASHE Standard 170-2021 Ventilation of Health Care Facilities*
  Contains a table of design parameters related to ventilation
  https://store.accuristech.com/standards/ashrae-170-2021?product_id=2212971
- *Facility Guideline Institute's Guidelines for Design and Construction of Outpatient Facilities 2022*.
  https://fgiguidelines.org/
- *ASHRAE's HVAC Design Manual for Hospitals and Clinics, Second Edition*
  Essential HVAC-oriented book on health care design.
  https://store.accuristech.com/standards/hvac-design-manual-for-hospitals-and-clinics-2nd-ed?product_id=1852624

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- *ASHRAE's Advanced Energy Design Guide for Small Hospitals – 30%*
  https://www.ashrae.org/technical-resources/aedgs/30-percent-aedg-free-download
- Technical Support Document: Development of the Advanced Energy Design Guide for Small Hospitals and Healthcare Facilities—30% Guide
  March 2010, NREL/TP-550-46314
  https://docs.nrel.gov/docs/fy10osti/46314.pdf
- *Community-Based Outpatient Clinics* – VA, 2014
  https://vatilms.va.gov/vatilms/reports/prototypes
- *PACT Space Module Design Guide* – VA, 2015
  https://vatilms.va.gov/vatilms/reports/PG-18-12%20Design%20Guides

Some additional data sources have been identified, but were not reviewed. The reasons that they were not reviewed were due to cost, the rawness of the data, or the time available:

- *SpaceMed Hospital and Healthcare Facility Planning Guidelines*
  Looks very useful, but costs $545
  https://www.spacemed.com/
- *State of U.S. Health Care Facility Infrastructure - ASHE*
  Might be useful – hard to know. Available to members
  https://www.ashe.org/facilityinfrastructure
- *California Department of Health Care Access and Information (HCAI) Facility Detail*, which includes healthcare in California, is a database that could be mined
  https://ams.aha.org/EWEB/DynamicPage.aspx?WebCode=ProdDetailAdd&ivd_prc_prd_key=a59b2c68-6db8-4bf5-b67c-8418a1fc5a26

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple databases lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple databases, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources.

OUTP4-Notes (n/a)

One interesting aspect revealed about this type of building, while performing web searches for additional data sources, is the large number of synonyms for this type of facility.

- ambulatory care
- ambulatory clinic
- ambulatory surgery
- clinic
- day hospital
- day patient
- day surgery
- freestanding birthing center
- freestanding outpatient diagnostic and treatment facility
- freestanding urgent care facility
- gastrointestinal endoscopy facility
- outpatient clinic
- outpatient facility
- outpatient surgery
- outpatient surgical facility
- partial hospital
- primary care outpatient facility
- same-day surgery
- small primary neighborhood outpatient facility
- small primary outpatient facility
- urgent care

OUTP5-Notes (B38)

The only source of information provided related to the typical size of an outpatient facility, from EIA CBECS 2018, indicates that the average size is 13,700 ft2, but the prototype is three times that size at 40,950 ft2. Please provide justification for the assumed size of the building.

OUTP6-Notes (B23)

Why is there no Construction Rate graph present? Usually, such a graph has been based on Dodge data for other prototypes.

OUTP7-Notes (B3)

Is any data taken from CEUS 2022 for outpatient buildings? If not, it should probably not be referenced.

OUTP8-Notes (n/a)

In general, for some of the other prototypes, the Notes tab has provided justification for many of the "big picture" assumptions, such as total floor area, number of stories, vintages, why certain years are chosen for representing vintages, and how similar buildings are differentiated.

OUTP9-Notes (A37:A38)

While this is a good description of an outpatient building, it would be good to indicate the exact boundaries assumed between data related to this building and other "adjacent" buildings such as medical office buildings, urgent care facilities, hospitals, birthing centers, dialysis centers, imaging centers, orthopedic centers, medical clinics, short term care facilities, long term care facilities. In some cases, "outpatient" may be defined as a medical facility where patients are present for anything less than 24 hours.

The FGI Guidelines for Design and Construction of Outpatient Facilities (2022), part 2, has the following subsections that indicate the range of subcategories for outpatient facilities:

- Common Elements for Outpatient Facilities
- Specific Requirements for General and Specialty Medical Services Facilities
- Specific Requirements for Outpatient Imaging Facilities
- Specific Requirements for Birth Centers
- Specific Requirements for Urgent Care Centers
- Specific Requirements for Infusion Centers
- Specific Requirements for Outpatient Surgery Facilities
- Specific Requirements for Freestanding Emergency Care Facilities
- Specific Requirements for Endoscopy Facilities
- Specific Requirements for Renal Dialysis Centers

- Specific Requirements for Outpatient Behavioral and Mental Health Centers
- Specific Requirements for Rehabilitation Therapy Facilities
- Specific Requirements for Mobile/Transportable Medical Units
- Specific Requirements for Dental Facilities

What is considered in and out of the scope of the prototype may impact the program of spaces and the fraction of each type of space in the building. Ultimately, it becomes a question of what is being modeled exactly and what it is intended to represent.

OUTP10-Notes (n/a)

No sources are cited that are specific to this type of building. It is not clear how any aspect of the model can be determined based on the lack of sources cited for this type of building.

OUTP11-Notes (C2:C5, C9:C14)

Provide links for each reference.

OUTP12-Notes (B19)

The PNNL 20405 report cited is from 2011, not 2010 as shown.

OUTP13-Prototype (A7:C16)

The shape, total floor area, aspect ratio, number of floors, roof type, window-to-wall ratio, shading geometry, floor-to-floor height, floor-to-ceiling height, and glazing sill height are all based exclusively on the PNNL. The cited source from PNNL being used is PNNL 20405 from 2011 titled “Achieving the 30% Goal: Energy and Cost Savings Analysis of ASHRAE Standard 90.1-2010” and for almost all the assumptions related to that prototype except for the 90.1 specific features, it references Deru 2011 which is “U.S. Department of Energy Commercial Reference Building Models of the National Building Stock” published by NREL (now NRL) as report number NREL/TP-5500-46861. That report, in turn, references GGHC 2007 and Doebber 2009. The second reference is “Development of the Advanced Energy Design Guide for Small Hospitals and Health Care Facilities – 30% Energy Savings,” NREL/TP-550-46314. This is the original source of the space layout and the source that is described as:

> “The drawings for a community hospital and surgery center were used to develop the prototype floor plans and space layouts. “

And

> “The first floor of the surgery center contains the critical care areas such as operating and recovery rooms; the second and third floors contain examination rooms, office spaces, and physical therapy areas.”

Where the surgery center is the term used in the report that is now referenced as “outpatient healthcare.” The floor layout diagrams are exactly the same as those shown on the “Zones” tab.

Two important points are:

- The source of the prototype should really reference the NREL 46861 report
- The prototype is based on one set of drawings for a surgery center.

This second point is especially interesting since one may assume that the nested references to PNNL and NLR reports provide legitimacy for a good prototype, but it is actually based on a single building that was designed for a specific use and was very likely not intended to be prototypical or representational of a category of buildings at all.

The lack of data used to derive information on this building type makes it difficult to assess how representative the outpatient healthcare prototype really is. It may be worth trying to gather data from California, regional, or national sources to see if this prototype is representative of the building category at all.

OUTP14-Prototype (A13:C13)

The model shows no shading geometry; this is very unlikely for the public entrances to a medical facility, which almost always have overhangs over the entrance to protect visitors from rain when being dropped off. This comment is essentially the same as HOSP6-Prototype.

OUTP15-Zones (A1:J121)

A spreadsheet that is consistent with the prototype’s origin (see OUTP13 -Prototype) was compared to the Zones tab. The spreadsheet used was for new construction from this source:

https://data.openei.org/submissions/162

The comparison of the Area Fraction, Area, Volume, and Multipliers was all consistent with the original.

The number of people was expected to be different in each row, based on different sources (Title 24 vs ASHRAE Standard 62.1), and was quite a bit lower at 329 total people instead of 399 in the original scorecard spreadsheet. This is still not unexpected.

Which zones were conditioned is quite different and not as expected. Every zone is shown as conditioned, including ones that were shown as not conditioned in the original spreadsheet. The list of zones that are shown as conditioned and were not originally included is:

- Floor 1 Electrical Room
- Floor 1 Elevator Pump Room
- Floor 1 Lobby Toilet
- Floor 1 MRI Toilet
- Floor 1 Nurse Janitor
- Floor 1 Nurse Toilet
- Floor 1 Pre-Op Toilet
- Floor 1 Storage
- Floor 1 Utility Janitor
- Floor 1 Utility Room
- Floor 2 Conference Toilet
- Floor 2 Janitor
- Floor 2 Reception Toilet
- Floor 2 Storage 1
- Floor 2 Storage 2
- Floor 2 Storage 3
- Floor 2 Utility
- Floor 2 Work Toilet
- Floor 3 Janitor
- Floor 3 Lounge Toilet
- Floor 3 Mechanical
- Floor 3 Office Toilet
- Floor 3 Physical Therapy Toilet
- Floor 3 Storage 1
- Floor 3 Storage 2

- Floor 3 Undeveloped 1
- Floor 3 Undeveloped 2
- Floor 3 Utility
- NE Stair
- NW Elevator
- NW Stair
- SW Stair

Some of these, such as the various toilets, may make sense to consider them conditioned, but many of the remaining zones may be more accurately assumed to be unconditioned, such as:

- Utility rooms
- Storage spaces
- Undeveloped spaces
- Janitorial
- Stairs

OUTP16-Zones (I)

The number of people column often shows zero since it is formatted to not show any digits after the decimal. This makes zones with fewer than 0.5 people show as 0. It should be formatted to show one digit after the decimal place. It is questionable whether to use non-integer numbers of people in models since they are inherently incorrect, but if they are used, they should be shown with at least one digit after the decimal point.

OUTP17-Zones (C2:C94)

Three air handling units are presumed in the scorecard, one for each floor. This seems problematic since rooms that may have special ventilation and temperature requirements are served by the same HVAC system. It seems very unlikely that a single system would serve the café and an operating room. This issue is mostly focused on the first level.

On the second level, both exam rooms and office spaces are served by the same HVAC system, which may occur, although this is not good design practice. The third level with office and physical therapy spaces seems the most reasonable.

In general, the quantity of HVAC systems seems too low for the type of facility, which may have complicated ventilation requirements and possibly pressurization requirements.

OUTP18-Zones (n/a)

The VA has a series of prototypes for Community Based Outpatient Clinic (CBOC), depending on the size of the core part of the facility that they call the Patient Aligned Care Team (PACT) Module. These are not building energy modeling prototypes but are typical design prototypes since they build many similar facilities. See below:

https://www.cfm.va.gov/til/Prototype/CBOC/CBOCPrototype-03-PFD.pdf

or

https://vatilms.va.gov/vatilms/reports/prototypes

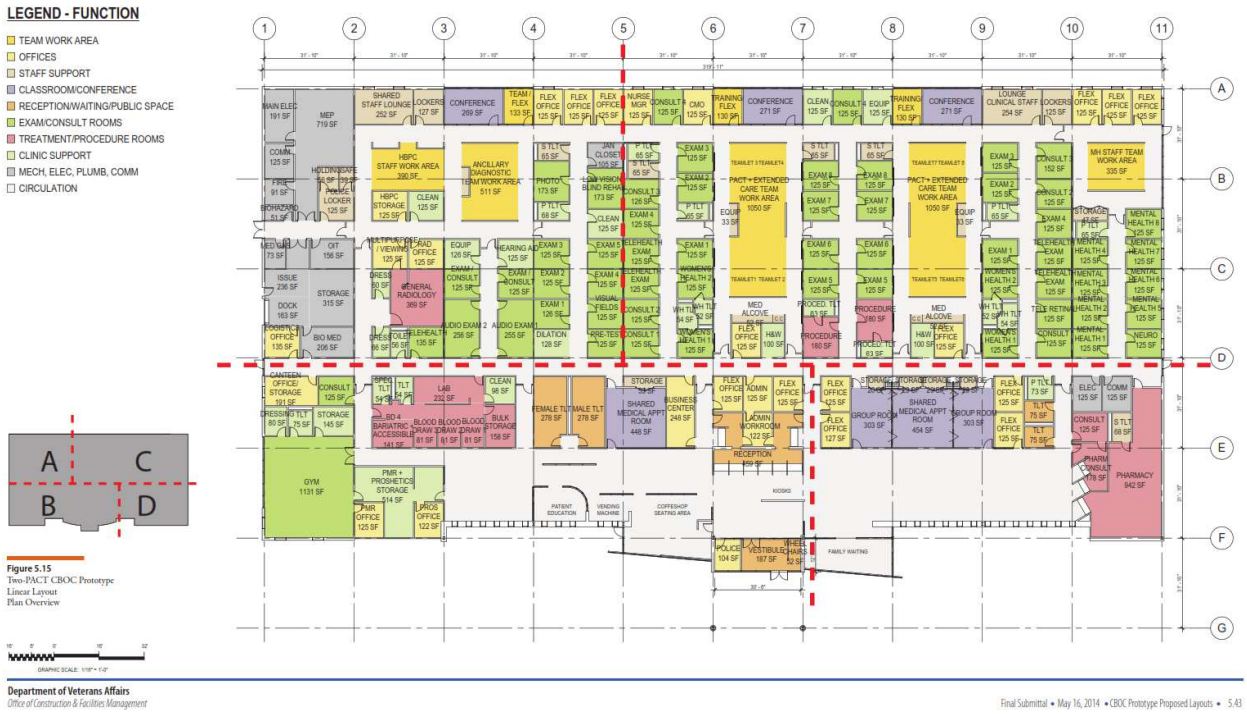
They have space programs for one, two, or three PACT designs, which are 17,000 ft2, 56,000 ft2, or 92,000 ft2. The closest for the Outpatient would be the two PACT designs, with the overview shown below:

VA CBOC PROTOTYPE (Two PACT)

Department	PFD Net Area	PFD Gross Area
Logistics	870	1,322
Patient Aligned Care Team (PACT) Module #1	6,220	9,454
Patient Aligned Care Team (PACT) Module #2	6,095	9,264
Audiology and Speech Pathology	1,585	2,409
Canteen	925	1,406
Home-Based Primary Care	480	730
Clinic Management	250	380
Engineering	200	304
Eye Clinic	1,910	2,903
Lobby	1,530	2,326
Mental Health	2,830	4,302
Pathology and Laboratory Medicine	860	1,307
Pharmacy	1,145	1,740
Physical Medicine and Rehabilitation	1,210	1,839
Police and Security	345	524
Prosthetics and Sensory Aids	415	631
Radiology	770	1,170
Ancillary Diagnostic Services	1,550	2,356
Business Services	360	547
Sum of Departments	29,550	44,916
Building Gross Factor		1.25
TOTAL BGSF		56,145

Where the PACT modules are mostly exam rooms and staff support areas.

An image of the overall two-PACT Community Based Output Clinic Prototype is shown:



Since this is not intended to be a "surgery center" (see OUTPx -Prototype), the specific space program is quite different and focuses on exam rooms and support for those rooms. It does include some types of spaces missing from the output patient prototype due to this assumption, such as gym, lab, blood draw, audio exam rooms, women's health rooms, low-vision rehab, mental health rooms, telehealth rooms, pharmacy, and others. In comparison, the output patient prototype based on the surgery center includes operating rooms, recovery rooms, anesthesia, pre-op rooms, and others. While no direct comparison is possible, it does illustrate that the exact purpose of a prototype can have a significant impact on the space plan and allocation.

This comparison also results in the question of whether a surgery center or an outpatient clinic (or perhaps urgent care) is more representative of the overall group of buildings that is trying to be captured by this prototype. This decision may have significant energy impacts since special conditioning (lower thermostat and controlled humidity) is needed for operating rooms that may or may not be present in all the buildings represented by the classification of "outpatient." Hopefully, a source of data to shed light on this question could be identified with further study.

OUTP19-Zones (n/a)

By space type, the outpatient facility prototype has these total areas:

Space Type	Area Fraction	Area (ft^2)
Breakroom/Lounge	2.9%	1199
Conference/Meeting	0.8%	336
Corridor Area	19.2%	7879
Dining Area (Cafeteria/Fast Food)	1.0%	420
Elevator	0.3%	140
Healthcare Facility and Hospitals (Exam/Treatment Room)	11.2%	4606
Healthcare Facility and Hospitals (Imaging Room)	3.7%	1508
Healthcare Facility and Hospitals (Medical Supply Room)	1.9%	765
Healthcare Facility and Hospitals (Nurse's Station)	1.4%	591
Healthcare Facility and Hospitals (Operating Room)	3.5%	1416
Healthcare Facility and Hospitals (Patient Room)	1.3%	527
Healthcare Facility and Hospitals (Physical Therapy Room)	4.6%	1892
Healthcare Facility and Hospitals (Recovery Room)	2.3%	948
Kitchen/Food Preparation Area	0.4%	182
Lobby, Main Entry	1.5%	622
Locker Room	1.9%	780
Mech/Elec	1.3%	539
Office Area (<250 square feet)	1.7%	676
Office Area (>250 square feet)	17.1%	7009
Reception area	3.6%	1493
Restroom	1.9%	790
Stair	1.1%	456
Storage	15.1%	6172
TOTAL		40946
Health Facility and Hospital Subtotal	29.9%	12253

The 29.9% for the actual "health" facilities seems low, but when considering that storage is 15%, office areas are 19%, and corridors are 19%, it is reasonable. Over half of the storage area is actually classified as undeveloped. The large fraction of the area devoted to office space, especially on the third and fourth floors, is difficult to understand. It would be beneficial to find a data source from which statistics can be drawn to validate this space allocation.

OUTP20-Zones (J)

The People (persons/1,000 ft2) column shows 10 people/1,000 ft2 for all the "Healthcare Facility and Hospital" spaces but that does not appear in Title 24-2025 Table 120.1A. It is

not clear where this number comes from. If it is from ASHRAE Standard 170, that standard should be cited.

OUTP21-Zones (J)

The People (persons/1,000 ft2) column shows 3 people/1,000 ft2 for all the “Mech/Elec” spaces, but that does not appear in Title 24-2025 Table 120.1A. It is not clear where this number comes from. I think typically zero is used for this value. Zero was used for other prototypes for mechanical and electrical rooms.

OUTP22-Zones (J)

The People (persons/1,000 ft2) column shows 5 people/1,000 ft2 for all the restrooms, but that does not appear in Title 24-2025 Table 120.1A. It is not clear where this number comes from. I think typically zero is used for this value since the ventilation for restrooms is driven by exhaust, and it is unusual to show scheduled occupancy for a restroom. Zero was used for other prototypes.

OUTP23-Zones (J)

The People (persons/1,000 ft2) column shows 20 people/1,000 ft2 for the locker rooms, but that does not appear in Title 24-2025 Table 120.1A. It is not clear where this number comes from.

OUTP24-Envelope (D2:D22)

The level of detail for the construction description was not as detailed as the hospital prototype.

OUTP25-Envelope (C3:D7)

Using a lightweight construction of “Metal frame + insulation + gypsum board” might be appropriate for an outpatient building, but it may be just as common to use concrete walls (either tilt-up or block). It would be good to find a source to justify this construction.

OUTP26-Envelope (R8:R9)

The roof u-value requirements for the "pre 1978" and "1978-1997" vintages are shown generally as the same as the values in the "1998-2007" vintage, with the note "Considering 30 years for roof/ insulation replacement." This is consistent except for CZ10, which showed 0.057 for the "1998-2007" vintage value and 0.78 for the "pre 1978" and "1978-1997" vintages. Since the values in other climate zones are 0.78, this is likely a copy-paste error. This is the same comment as NURS36-Envelope, which implies that the table was copied from that spreadsheet.

OUTP27-Envelope (H4)

The requirements for the 1978-1997 vintage for exterior walls are described as using Title 24-1986 and "Package A with heat capacity around 10 (Btu/F/ft^2)", and the values are the same as in the small hotel scorecard, but elaborate further:

> "Copied from High-rise office alternative requirements (Table 2-53W): Package A with heat capacity between 10-14.99 (Btu/F-ft^2) *T24, 1986, Table 2-53BB presents U-factors less than the next vintage (for wall density of 26-40 lb/ft^2 & Degree day<3500)!"

If this is the same source, please include this additional detail.

OUTP28-Envelope (H13:X14, H18:X19)

The pre-1978 and 1978-1997 vintages for fenestration are described as being based on

> ASHRAE Handbook of Fundamentals (Table 4-Ch15) for single pane & T24, 2013 for double pane (88%-12%)*
>
> ASHRAE Handbook of Fundamentals (Table 10-Ch15) for single pane & T24, 2013 for double pane (88%-12%)*

For U-Value and SHGC, respectively.

The formulas used assume a U-value of 1.12 and an SHGC of 0.79 for the single pane. Based on the cited source of the ASHRAE Handbook of Fundamentals, these correspond to "Fixed" and "Aluminum" frames "Without Thermal Breaks." It would be good to add this level of detail to the description. Also, the exact year of the ASHRAE Handbook of Fundamentals should be included.

The “88%-12%” ratio used is reflected in the formulas, but no basis for this ratio is provided. Please add one to the description. The asterisk implies that a citation was expected to be added but was not.

OUTP29-Envelope (n/a)

Several envelope table issues were reported for the small hotel that also apply to the hospital prototype table:

- SHSC32 – roof replacement should be weighted
- SHSC35 – should say “11%-20%” not “10%-20%”
- SHSC37 – use non-north for all windows
- SHSC37 – consider weighting all standards for historic values
- SHSC39 – shouldn’t average single and double pane window criteria since non-physical

For brevity, they will not be repeated. This is very similar to comment HOSP23.

OUTP30-Envelope (E25:X33)

To verify the infiltration values, the four and six side areas were based on the values from the Prototype tab:

- Area 4 sides = 16,716 ft2 (based on above-grade wall area)
- Area 6 sides = 16,716 ft2 + 2 x 14,779 ft2 (based on roof area) = 46,274 ft2
- Ratio of 6 sides to 4 sides = 46,274 / 16,716 = 2.77

So, according to the referenced report

$$I_{design}\text{ at the building envelope } = 0.112 \times I_{75}\text{ at building envelope} \qquad 2$$

$$I_{design\ (4-side)} = I_{design\ (5-or\ 6-side)} \times \left(\frac{\text{Total leakage area } (5-\text{ or } 6-\text{ side})}{\text{Exterior Wall area}}\right) \qquad \textit{Equation 3}$$

Based on this, the values shown in the table are correct.

OUTP31-HVAC System (C16)

The footnote that says:

> * Et = Combustion efficiency

but this should probably be for “Ec.”

OUTP32-HVAC System (D10:D12)

The cell for the pre-1978, 1978-1997, and 1998-2007 vintage for gas furnace says:

> Gas-fired forced air furnaces with input ratings
>
> ≥225,000 Btu/h--75% of the input rating

But no source for that value is provided, and it seems lower than expected. Perhaps 80% thermal efficiency is what should be used.

OUTP33-HVAC System (D13)

The 2008-2018 vintage using Title 24-2013 for the gas furnace shows:

> Warm-Air Duct
>
> Furnaces, Gas-Fired-All Capacities--80% Ec

But this should be 80% Et, thermal efficiency, not combustion, according to Table 110.2J of Title 24-2013.

OUTP34-HVAC System (F13)

The Source cell for the 2008-2018 vintage using Title 24-2013 shows:

> T24, 2013, Table 110.2-D (for DX Coil) -Table 110.2-I (for Gas Furnace thermal efficiency)

But they should be referring to Table 110.2-A and Table 110.2-J.

OUTP35-HVAC System (D14:F14)

For new construction, the requirements of Title 24-2025 do not contain requirements for gas furnaces or gas boilers. The values shown on the table are from Title 24-2013. Shown below

> Warm-Air Duct Furnaces, Gas-Fired-All Capacities--80% Ec
>
> Gas-Fired Boiler ≥ 300,000 Btu/h and ≤ 2,500,000 Btu/h-- 84% Et
>
> T24, 2013, Table 110.2-A (for DX Coil) - Table 110.2-J (for Boiler thermal efficiency)- Table 110.2-I (for Gas Furnace thermal efficiency)

Given the focus on electrification, shouldn't a heat pump-based system be used instead?

OUTP36-HVAC System (F21, F24:F26)

A fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction. This is the same comment as LHSC52 and OFFM43.

OUTP37-Thermostat (A2:S6)

Several special spaces should have unique thermostat settings.

- Operating rooms are often kept quite cool in the range of 60°F to 65°F
- Imaging often depends on the requirements of the equipment and patients, and is sometimes also kept very cool

This is not an exhaustive list; other special spaces may have different thermostat settings. The Note describes some special setpoints that the PNNL prototype used. This is a similar comment to HOSP33.

OUTP38-Ventilation (F)

For three of the space types, there is no cfm/ft2 value in Table 120.1-A of Title 24-2025: Mech/Elec, Restroom, and Locker Room. These show 0.15 in column F but should probably be shown as "n/a." See OUTP21, OUTP22, and OUTP23.

OUTP39-Ventilation (G)

The “Healthcare Facility and Hospitals” spaces in the column labeled “Required Exhaust Rate (cfm/ft^2) [T24, 2025, Table 120.1-B – minimum exhaust rates]” contain values that are not in the source.

- Healthcare Facility and Hospitals (Exam/Treatment Room)
- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Imaging Room)

The values appear to be calculated instead of being from the table. For most of these spaces, a note states, “Exhaust Rate assumed according to PNNL outpatient model,” but that should be elaborated as to why they are present when not based on a code requirement.

OUTP40-Ventilation (I2)

The column label “Min OA Ventilation Requirement (cfm/ft^2) [Table 4-A, 2022 California Mechanical Code]” is a little misleading since it contains computed numbers rather than the values from the source. The source uses a minimum outdoor ACH (air change rate) of 2 for most health care spaces, which translates to 0.33 cfm/ft2 (based on a 10 ft ceiling).

OUTP41-Ventilation (I32:I34)

The three operating rooms are shown with an incorrect minimum outside air ventilation requirement. They are shown using 0.33 cfm/ft2, which corresponds to two air changes per hour according to the source, Table 4-A of the 2022 California Mechanical Code, but that code shows that four air changes per hour should be used for operating rooms. This directly impacts the total CFM computation for these spaces.

OUTP42-Ventilation (L)

The “Total cfm Driven by Exhaust” column only has one value for the first row, and then the corresponding cell for other rows that have values for the required exhaust rate (columns G and H) shows zero. This means the ventilation rates for spaces driven by exhaust rates are

zero or are based on the ventilation based on the number of people or floor area instead. This is an error that definitely applies to:

- Restrooms
- Locker rooms
- Kitchens

It may also apply to the following spaces, which also show an exhaust rate, although no source for the values was found.

- Healthcare Facility and Hospitals (Exam/Treatment Room)
- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Medical Supply Room)
- Healthcare Facility and Hospitals (Imaging Room)

Also see OUTP39.

OUTP43-Ventilation (135:178)

The background data shown in these rows is for Table 4-A of the California mechanical code and is quite useful and perhaps should not be deleted.

OUTP44-Ventilation (A182:I206)

The hidden table showing outdoor airflow rate by space type is a very useful summary and should not be deleted or hidden.

OUTP45-Water Heater (D39)

The water heater capacity (Btu/hr) is based on the heat output times the water heater system factor of 0.6 as described in the “ACM 2025 5.9.2 rated capacity section,” and that is done for the gas and total capacity, but for the electric resistance water heater, a water heater system factor of 0.85 appears to be used. No source is provided for this, so it may be a mistake.

OUTP46-Water Heater (n/a)

Missing space types for laundry and special sanitation, if present, would require additional water heaters to provide higher temperature water (similar to HOSP36)

OUTP47-Water Heater (B43)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as LHSC64-Water Heater, SHSC61-Water Heater, and HOSP39-Water Heater.

OUTP48-Interior Lights (n/a)

The following space types allow additional lighting power, and it is very likely that for many of these space types, the additional lighting power would be used:

- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Nurse's Station)
- Healthcare Facility and Hospitals (Recovery Room)
- Healthcare Facility and Hospitals (Physical Therapy Room)
- Healthcare Facility and Hospitals (Patient Room)
- Dining Area (Cafeteria/Fast Food)
- Office Area (<250 square feet)
- Office Area (>250 square feet)
- Conference/Meeting
- Lobby, Main Entry
- Corridor Area
- Restroom
- Breakroom/Lounge
- Stair
- Mech/Elec

A slightly different list applies to existing buildings, additional lighting power from T24-2019. This is a similar comment to HOSP40.

OUTP49-Interior Lights (B4:24 E4:24)

The lighting power densities for new construction (T24-2025) and existing (T24-2019) were checked and are ok.

OUTP50-Interior Lights (G4:G16)

No justification is used for using T24-2019 for existing lighting power densities. It seems a bit optimistic for many of the space types in an outpatient facility to have their lighting systems replaced that quickly. Similar comment to HOSP42.

OUTP51-Ext Lights (A9:B9)

The parking area estimate states that it is based on "400 ft^2 for every parking space, one parking space per 830 ft^2 of gross building area," and the source is "2019 Outdoor Lighting CASE Report & PNNL 90.1-2010 TSD," but the value of 830 sqft from the CASE report is for warehouses:

Table 38: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for General Hardscape – Table 140.7-A

Assumptions for Statewide Estimates - General Hardscape		
General Hardscape	**Assumptions**	**Area Multipliers for Construction S.F.**
for Large Office, Small Office, Food, Restaurant, College	1 parking space per 250 sf of gross building area	1
for Hotel, Retail, School, Other	1 parking space per 360 sf of gross building area	0.7
for NR Warehouse, Ref. Warehouse	1 parking space per 830 sf of gross building area	0.3

A possible alternative way of computing this can be found here:

https://blog.spacemed.com/estimating-healthcare-facility-parking-requirements/

This is very similar to the comment HOSP45.

OUTP52-Ext Lights (A11:B11)

The basis for the ratio of 0.1 ft^2 per 1 ft^2 of gross building area for hardscape ornamental lighting comes from the referenced 2019 outdoor lighting report, but the value does not seem to apply to health facilities, according to the table where it appears:

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications										
		Applied to % of Building S.F. in Category								
Lighting Allowance	**Assumptions**	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Building Entrances or Exits	**1 per 5000 sf of building interior** (20 occupants per door, 250 occ/sf)	100%	100%	100%	100%	100%	100%	100%	100%	99%
Primary Entrances to Senior Care Facilities, Police Stations, Hospitals, Fire Stations, and Emergency Vehicle Facilities	**1 per 5000 SF of gross building area** (1 primary entrance per building)									1%
Drive Up Windows	**1 per 1500 SF of gross building area** (2 locations per building; 1000 sf building)			30%						
Vehicle Service Station Uncovered Fuel Dispenser	**1 per 100 sf of gross building area** (1 fuel dispenser face per 25 sf of station building interior)									0.01%
Automated Teller Machines	**400W MH luminaire as typical standard practice, switch to 250W limit for first location,** 2500 sf per ATM installation.									1%
Outdoor Sales Frontage	**0.2 LF per sf of gross building area** (1 display parking space per 50 sf of building interior)									1.5%
Hardscape Ornamental Lighting	**0.1 SF per SF of gross building area**	50%	50%	50%	25%		50%	25%	25%	5%
Building Facades	**30' building height, 2 floors per building** (20% of applicable facades are lit)	25%	50%	50%	25%		50%	25%	25%	5%
Outdoor Sales Lots	**4 SF of sales lot per sf of gross building area** (1 display parking space per 50 sf of building interior)									1.5%
Vehicle Service Station Hardscape	**11 SF per SF of gross building area**									1%
Vehicle Service Station Canopies	**1.2 SF of canopy per SF of gross building area**									1%
Sales Canopies	**0.1 SF of canopy per SF of gross building area**									5%
Non-sales Canopies	**0.1 SF of canopy per SF of gross building area**	25%	25%	25%	25%		25%	25%	25%	5%
Guard Stations	**0.00043 sf per SF of gross building area** (1 12x18 guard station per 500,000 sf of total construction)	100%				100%			100%	100%
Student Pick-up/Drop-off zone	**0.0173 sf per SF of gross building area** (1 12x72 drop off per 50,000 sf of total construction)							100%		
Outdoor Dining	**1 sf per 5 sf of gross building area** (20% of typical building sf)		2.5%	50%	2.5%					
Special Security Lighting for Retail Parking and Pedestrian Hardscape	**1 SF per 100 SF gross building SF** (1% of hardscape)		100%	100%	100%					50%

OUTP53-Ext Lights (A12:C12, A21:B21)

For building facades, the value of 10,020 ft2 is used, but it is not clear where this number comes from. The formula for it points to the perimeter of the building, but that cell uses the value of 10,020, so it seems like a circular calculation. Where is the value from?

OUTP54-Ext Lights (A12:G12)

The building façade area assumes a factor of 0.5 of the perimeter walls; less than two stories are lit, but Table 36 from the CASE report suggests using 5% for "other" buildings that are not on the list. Same as HOSP46.

OUTP55-Ext Lights (C13)

The area of the non-sales canopy is 1365 ft2, but no source for that is provided. While it is inconsistent with the assumption on the prototype tab showing no shading, it is probably appropriate to consider an entrance canopy to exist on this building type.

OUTP56-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72-Ext Lights, SHSC67-Ext Lights, and HOSP44.

OUTP57-Equipment (A2:23)

Several of the values shown for plug load power density are not consistent with the values from T24-2022 Appendix 5.4A, including office area, nurses' station, operating room, and imaging room. For the operating room and imaging room, the 5.4A values do not make sense. It might be more appropriate to keep the 5.4A values for office and nurses' station to be consistent. This is the same comment as HOSP50.

OUTP58-Equipment (A7:F7)

For operating rooms, 11 W/sqft is assumed. The value from T24-2022 Appendix 5.4A is 1.5 W/sqft, which is clearly too low. The AEDG 50% TSD uses 4 W/sqft (1 W/sqft plug load and 3 W/sqft process loads). The Target 100 analysis includes data on actually monitoring operating rooms. This is the same comment as HOSP51.

OUTP59-Equipment (A5:F5)

For imaging rooms, 10 W/sqft is assumed. The value from T24-2022 Appendix 5.4A is 1.5 W/sqft, which is clearly too low. The AEDG 50% TSD uses 9 W/sqft (1 W/sqft plug load and 8 W/sqft process loads). The Target 100 analysis includes data on actually monitoring operating rooms. This is the same comment as HOSP52.

OUTP60-Equipment (A33:E36, C28:C29)

Three elevators are assumed in the prototype, which seems reasonable for the size and requirements of this building, but two may be sufficient. Unfortunately, the use of three elevators is justified based on the office building estimation approach in the article:

https://buildingandinteriors.com/number-of-elevators-required-in-a-building/

which states:

> One elevator for approximately 50,000 square feet is in use. For every two floors or two and a half floors, consider adding one more elevator.

These rules seem to justify two elevators, not three, as shown in the calculations. This may be due to the complicated formula in cell C36.

The same article provides some guidance for medical buildings, which is almost identical to another article at:

https://resoluteelevator.com/elevators-per-building/

Includes the following for medical facilities:

- In buildings with significant staff and visitor traffic, consider separate passenger elevators for staff and patients.
- Two patient elevators at a minimum for multi-story medical facilities, and one additional elevator for every 100 beds.
- Additional elevators may be required if visitor areas are on higher floors. This is often dependent on the location of the medical facility and overall travel distance.
- Dirty elevators may be required or needed. Moving medical waste sometimes requires a special elevator.
- Additional elevators may be required if operating areas, cafeterias, laundry, central supplies, etc., are on upper levels.

- Consider what each area of the medical facility is doing and where each elevator stops.
- You don't necessarily need a hospital bed-sized elevators throughout the facility.

In addition, some states may have rules about the number of elevators just based on occupancy and not building type

https://www.revisor.mn.gov/rules/4660.6600/

Which states:

> The number of elevators required are: one elevator for one to 59 persons above the first floor, two elevators for 60 to 200 persons above the first floor, and three elevators for 201 to 350 persons above the first floor. One additional elevator shall be added for each 150 persons in facilities with more than 350 persons above first floor.

The prototype shows 190 people above the first floor.

Overall, the quantity of elevator assumptions should be examined carefully.

OUTP61-Equipment (A29:I29)

The referenced requirement in Title 24-2025 §120.6 (f) matches most of the descriptions, but that section has a broad exception:

> Exception to Section 120.6(f): Elevators located in healthcare facilities.

Since this facility would qualify for that exception, it is not clear if this is a valid set of assumptions to use.

OUTP62-Equipment (D29)

The lighting shut-off requirement for elevators of "Automatic shut-off after ≤ 5 min no-occupancy" shown in this cell is not consistent with the referenced Title 24-2025 §120.6 (f), which shows 15 minutes, not 5 minutes. Also note, this is stated in cell E29.

OUTP63-Equipment (H28:H29, B42:D42, B46:D46)

The large difference in the design load lift for existing and new construction elevators is not substantiated, but shows about a 45% reduction in design load.

OUTP64-Schedule (C)

The levels that apply to the schedules seem to be reversed. The health-related levels are primarily the first and second levels, and the offices are primarily on level three, based on the descriptions on the Zone tab. The description in this column shows the office on level 1 and health on levels 2 and 3.

OUTP65-Schedule (AB30)

The schedules referencing “T24 2022 - Appendix 5.4B” were all verified and found to match, except one number for hour 24 for weekday office occupancy, which shows 0 but should be 0.05. Probably a typo.

OUTP66-Schedule (F37:W37)

The schedule for office equipment is based on “plug load and schedules project with CEC, 2025,” which is assumed to be the study titled “Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual” from August 7, 2025. The values for Saturday are 0.43 for all hours, but the prototype shows values from 0.2 to 0.9 depending on the hour.